

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - n79 (4.4-5.0 GHz 5G NR) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting n79 (4.4-5.0 GHz 5G NR) for Millimeter-Wave Beamforming. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 937.0 MHz and bandwidth of 163.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: n79 (4.4-5.0 GHz 5G NR)
- Application: Millimeter-Wave Beamforming
- Substrate: 36° YX LiTaO3
- Package: Fan-Out WLP
- Design Tool: PathWave EM Design
- Design Center: Osaka Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting n79 (4.4-5.0 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	937.0 MHz
Bandwidth (3 dB):	163.9 MHz
Insertion Loss Target:	<= 1.7 dB
Return Loss Target:	>= 14.0 dB
Out-of-Band Rejection:	>= 50.0 dB
Isolation (Duplexer):	>= 31.0 dB
Q Factor:	2708
Temperature Coefficient:	-18.9 ppm/°C
Die Size:	0.6 x 1.4 mm
Wafer Size:	6-inch
Number of Resonators:	6
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

- 1. Open PathWave EM Design and load the BAW (Bulk Acoustic Wave) template project.
- 2. Define the substrate stack: 36° YX LiTaO3 with specified layer thicknesses.
- 3. Set design targets: center freq 937.0 MHz, BW 163.9 MHz.
- 4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
- 5. Optimize resonator geometry using gradient descent (target IL < 1.7 dB).
- 6. Verify coupling coefficients between resonator stages.
- 7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
- 8. Export GDS-II layout for mask fabrication.
- 9. Fabricate prototype wafers (6-inch, 36° YX LiTaO3).
- 10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
- 11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 2811 MHz.
- 12. Compare measured results against simulation and specification limits.
- 13. Perform temperature cycling (-40°C to +85°C) and re-measure.
- 14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.7 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 246 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 2.2 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2024-09-14
Devices Tested: 42
Insertion Loss (meas): 1.39 dB
Return Loss (meas): 16.4 dB
Rejection (meas): 52.7 dB
Isolation (meas): 36.0 dB
Q Factor (meas): 2708
Group Delay Variation: 7.8 ns
Power Handling: +31 dBm
Die Yield: 88.4%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated below-target performance for n79 (4.4-5.0 GHz 5G NR).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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